

Assignment 1

Part I:

Q1) A traditional file system stores data in separate files, whereas a DBMS stores data in a structured database and provides features such as security, data consistency, and efficient querying. For example, in a university student management system, information about students, courses and grades must be accessed and updated by multiple departments. A DBMS ensures that all users see consistent data and prevents duplication or loss of information, making it a more reliable choice.

Q2) A strong entity can be uniquely identified by its own primary key, while a weak entity cannot be uniquely identified using only its own attributes and depends on a strong entity. The primary key of a weak entity is formed by combining its partial key with the primary key of the strong entity. The relationship between a weak entity and its owner strong entity is called an identifying relationship because this helps uniquely identify the weak entity

Q3) Phone number should be modeled as a multivalued attribute because a student may have more than one phone number. Therefore, multiple phone numbers can be associated with a single student.

Q4) Candidate key: A smallest set of attributes that uniquely identifies a record. Super key: Any set of attributes that uniquely identifies a record. It contains candidate key and extra attributes. From the relation roll_no, email are candidate keys. Super keys that are not candidate keys: (roll_no, name) and (email, branch)

Q5) Foreign Key Constraint is violated in this case. This constraint prevents records from referencing non-existing records on another table and maintains data consistency. The insert operation should be rejected until a valid student_id is provided.

Q6) A junction table (associative table) must be created as additional structure. The primary key consists of primary keys of the participating entities usually as a composite key. The relationship not simply be represented as a column in one of the participating entity tables because a many-to-many relationship allows multiple records on both side, which cannot be represented by a single column without causing redundancy and data inconsistency.

Part II:

Part A:

Entities and Key attributes

1. Customer

- CustomerID (Primary Key)
- Name
- Email
- Phone_number

2. Restaurant

- RestaurantID (Primary Key)
- Name
- Address

3. Order

- OrderID (Primary Key)
- OrderDate

4. MenuItem

- MenuItemID (Primary Key)
- ItemName
- Price

Relationships

1. Customer places order
Cardinality: 1:N
One customer can place multiple orders, but each order is placed by only one customer
2. Restaurant receives order
Cardinality: 1:N
A restaurant can receive many order, but each order is associated with exactly one restaurant
3. Restaurant offers MenuItem
Cardinality: 1:N
A restaurant can offer multiple menu items, while each menu item belongs to one restaurant
4. Order contains MenuItem
Cardinality: M:N
An order may contain multiple menu items, and the same menu item can appear in many different orders

Part B:

1. Multivalued attribute:
Customer(phone_number), a customer can have multiple phone numbers, so it can have multiple values for a single customer.
2. Composite attribute:
Restaurant(Address), an address can be divided into street, city, state and PIN code making it a composite attribute

Part C:

Relational schema: OrderItem(OrderID, MenuItemID, Quantity)

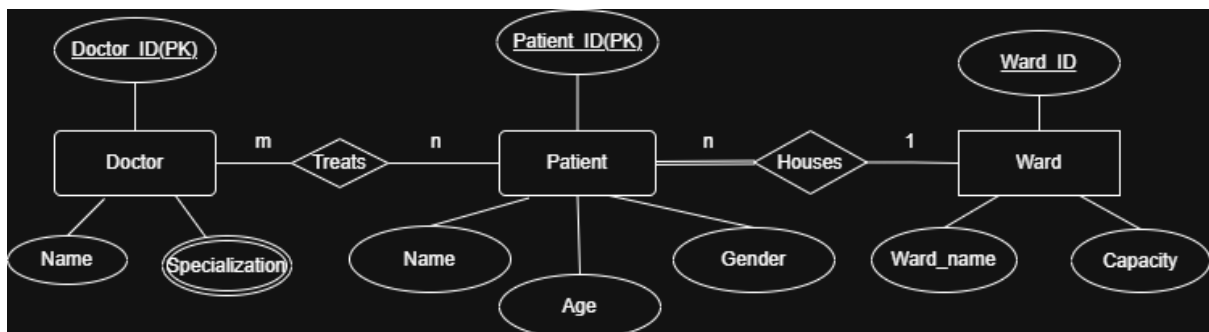
Primary keys: OrderID, MenuItemID

Foreign keys: OderID from Order(OrderID), MenuItemID from MenuItem(MenuItemID)

Additional attributes: Quantity, stores number of units of each menu item are included in an order

Part III:

Q1)



Q2) The Doctor–Patient relationship is many-to-many because a doctor can treat multiple patients and a patient can be treated by multiple doctors during their stay. This relationship can be represented using an associative entity called Treats. The composite primary key of Treatment would consist of DoctorID and PatientID. In the Houses relationship, Patient has total participation because every patient must be assigned to a ward, while Ward has partial participation because a ward can exist without patients. In the Treats relationship, both Doctor and Patient have partial participation because they may exist without being involved in treatment at a given time. One attribute design decision was modeling Specialization as a multivalued attribute because a doctor may have expertise in more than one medical field.